

Class 8: A Brief History of Wi-Fi

Protocols, PHY Waveforms, and the Design Choices from
802.11b to 802.11be

Xin Liu

Florida State University
xliu15@fsu.edu

CIS 4930/5930 Future Edge Networks
xinliulab.github.io/FSU-CIS4930-CIS5930-Future-Edge-Networks

Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz
Bandwidth	20 → 40 → 80 → 160 → 320 MHz

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz
Bandwidth	20 → 40 → 80 → 160 → 320 MHz
Constellation	BPSK → 64 / 256 / 1024 / 4096-QAM

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz
Bandwidth	20 → 40 → 80 → 160 → 320 MHz
Constellation	BPSK → 64 / 256 / 1024 / 4096-QAM
Space	SISO → SU-MIMO → MU-MIMO

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz
Bandwidth	20 → 40 → 80 → 160 → 320 MHz
Constellation	BPSK → 64 / 256 / 1024 / 4096-QAM
Space	SISO → SU-MIMO → MU-MIMO
Scheduling	one user → OFDMA resource units

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Wi-Fi grew by spending new resources



The icon never changed. The radio under it did — six ways.

Preview, not prerequisite — every row gets its own slide today. Keep a tally.

Dimension	How Wi-Fi spent it
Spectrum	2.4 → 5 → 6 → 60 GHz
Bandwidth	20 → 40 → 80 → 160 → 320 MHz
Constellation	BPSK → 64 / 256 / 1024 / 4096-QAM
Space	SISO → SU-MIMO → MU-MIMO
Scheduling	one user → OFDMA resource units
Links	one band → multi-link operation

Arrows list resources, not chronology: 60 GHz (2012) predates 6 GHz (2020s).

One family identity; each amendment spends a new dimension.

QAM = Quadrature Amplitude Modulation MIMO = Multiple Input, Multiple Output
OFDMA = Orthogonal Frequency-Division Multiple Access

IEEE 802.11 standard family; IEEE 802.11be; Cisco 802.11ax white paper



Amendment names vs. Wi-Fi generations

IEEE amendment	Market name	PHY shorthand
802.11b	Wi-Fi 1	HR-DSSS / CCK
802.11a	Wi-Fi 2	OFDM at 5 GHz
802.11g	Wi-Fi 3	OFDM at 2.4 GHz
802.11n	Wi-Fi 4	HT: MIMO + 40 MHz
802.11ac	Wi-Fi 5	VHT: wider + 256-QAM
802.11ax	Wi-Fi 6 / 6E	HE: OFDMA + 1024-QAM
802.11be	Wi-Fi 7	EHT: MLO + 320 MHz
802.11ad / ay	WiGig	directional 60 GHz

IEEE names amendments; the Wi-Fi Alliance names generations. Wi-Fi 1–3 are retroactive, informal labels.

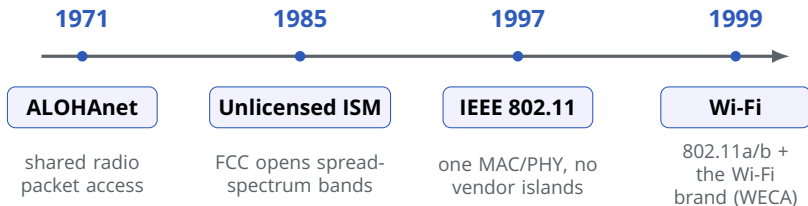
One radio, two naming systems: engineering (IEEE) and marketing (Wi-Fi Alliance). The orange row is a parallel branch.

HT/VHT/HE/EHT = High / Very High / High-Efficiency / Extremely High Throughput MLO = Multi-Link Operation

IEEE 802.11 interactive timeline; Keysight 802.11ad concepts



Four milestones made Wi-Fi possible



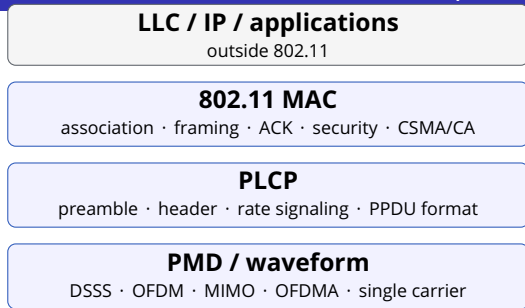
Q: Which of the four is regulatory rather than technical?

A: 1985. The FCC opened the “junk bands” microwave ovens pollute — nobody needed permission to ship a radio there, and Wi-Fi built an empire on that permissionless ground.

ISM = Industrial, Scientific, and Medical bands FCC = Federal Communications Commission

IEEE 802.11 interactive timeline; Wi-Fi Alliance (WECA) newsroom, first certifications April 2000

The 802.11 stack: stable MAC, moving PHY



**PHY: rebuilt
every generation**

PPDU = the PHY-layer transmission unit placed on air: preamble + header + payload.

Same family behavior above, different ways to turn bits into radio energy below. That split is Wi-Fi's longevity.

MAC = Medium Access Control PLCP = Physical Layer Convergence Procedure PMD = Physical Medium Dependent (1997 split; later revisions fold PLCP/PMD into one PHY)

IEEE Std 802.11 (MAC and PHY specifications)

1997 solved shared access before speed

Ethernet detects collisions mid-send. Why can't a radio?



DIFS (Distributed Inter-Frame Space) = a fixed polite pause after the channel goes quiet

Original PHY choices

- 1 or 2 Mb/s
- FHSS or DSSS at 2.4 GHz
- Infrared (never shipped)

Because a radio...

- Own TX deafens own RX
- Not every station is audible
- So: back off, then demand ACK

CSMA/CA avoids collisions; it cannot detect them while transmitting. Silence proves nothing — only an ACK does.

CSMA/CA = Carrier Sense Multiple Access / Collision Avoidance FHSS/DSSS = Frequency-Hopping / Direct-Sequence Spread Spectrum

IEEE Std 802.11-1997 baseline

802.11b: coded spreading at 2.4 GHz

11 Mb/s max PHY rate $\approx$ **22 MHz** channel **2.4 GHz** band

one symbol

→



a patterned chip sequence (conceptual; Barker = 11 chips, CCK = 8 complex chips)

- **1 / 2 Mb/s:** Barker spreading, as in 1997
- **5.5 / 11 Mb/s:** CCK — bits live in code-word phases
- Receiver knows the pattern — it digs the signal out of noise

$\approx$ 22 MHz-wide channels on 5 MHz centers is why
North America gets only three clean 2.4 GHz channels.

DSSS = Direct-Sequence Spread Spectrum CCK = Complementary Code Keying chip = one element of the spreading pattern processing gain = the technical name for that noise-digging

[IEEE 802.11 interactive timeline; IEEE Std 802.11](#)

802.11a: many orthogonal subcarriers

54 Mb/s max rate

20 MHz channel

5 GHz band



at each dashed sampling line, every neighbor is exactly zero (conceptual)

You met OFDM in Class 3 (LTE). Wi-Fi got there in 1999 — same idea, 20 MHz at a time.

- 64-point FFT
- BPSK up to 64-QAM
- 48 data + 4 pilot subcarriers
- 6–54 Mb/s rate set

Q: If OFDM was better, why did 802.11b win the first round?

A: 11a silicon shipped ~2 years later (Atheros, late 2001), cost more, crossed walls worse at 5 GHz, and could not talk to the installed 11b base. A great waveform loses to a cheap, entrenched ecosystem — for a while.

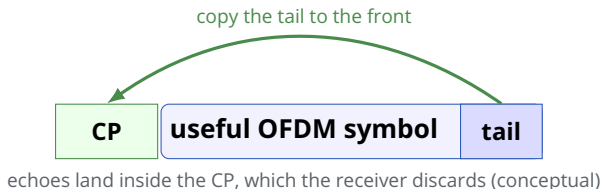
OFDM = Orthogonal Frequency-Division Multiplexing FFT = Fast Fourier Transform BPSK = Binary Phase-Shift Keying

[IEEE Std 802.11 \(OFDM PHY\); IEEE 802.11 timeline](#)



Why OFDM needs a cyclic prefix

Problem: indoor echoes smear one symbol into the next.



- Delay spread becomes a bounded guard region
- Each subcarrier sees one complex gain
- Equalization = one multiply per subcarrier

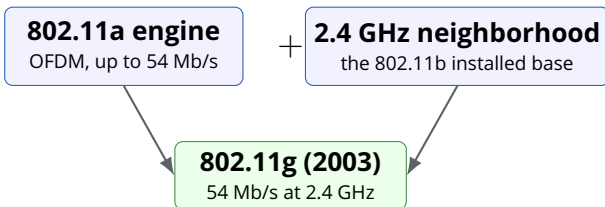
The cyclic prefix turns multipath smearing into a fixed, budgeted overhead — paid by every OFDM Wi-Fi since 11a.

CP = Cyclic Prefix — Wi-Fi calls this the guard interval (GI) delay spread = arrival-time spread of echoes

IEEE Std 802.11 (OFDM PHY, guard interval)



802.11g: the OFDM engine comes home



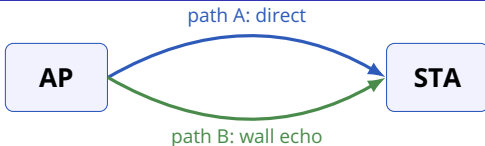
- Same OFDM waveform, more popular band
- Legacy 11b stations cannot decode OFDM
- So mixed networks send protection frames first

Backward compatibility bought adoption and cost airtime:
protection overhead is the rent paid to the installed base.

protection frame = a short legacy-format announcement that reserves the channel before OFDM speaks

[IEEE 802.11 interactive timeline; IEEE Std 802.11](#)

802.11n: multipath becomes an asset



more independent paths → more solvable equations → more streams (conceptual)

stream = an extra simultaneous data lane through the same channel and band.

- **40 MHz** channel bonding
- Up to **4 SU-MIMO** streams
- Short guard interval
- **A-MPDU** frame aggregation

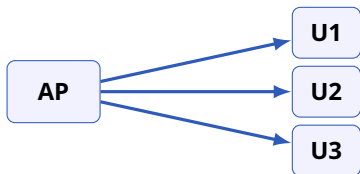
600 Mb/s ceiling = 4 streams × 40 MHz × 64-QAM 5/6
× short GI. Typical clients implemented far fewer streams.

SU-MIMO = Single-User MIMO A-MPDU = Aggregated MAC Protocol Data Unit STA = station (client) 5/6 = coding rate — the data fraction; the rest is error protection

[IEEE 802.11 interactive timeline; IEEE Std 802.11n-2009](#)

802.11ac: wider, tighter, more users

- **Wider:** up to 160 (80+80) MHz
- **Tighter:** 256-QAM (8 bits)
- **More streams:** 8 maximum
- **DL MU-MIMO:** Wave 2 gear



downlink MU-MIMO: one transmission, several spatially separated users

Q: Will any laptop ever see the 6.9 Gb/s on the box?

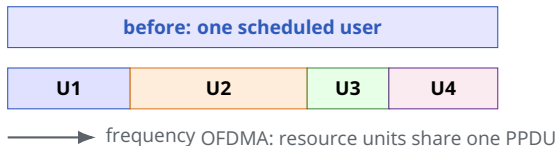
A: No. A maximum PHY rate is a configuration ceiling — every optimistic option at once — not expected application throughput.

MU-MIMO = Multi-User MIMO DL = downlink “Wave 2” = a product-certification wave, not a standards phase

Cisco, Migrating to 802.11ac; Cisco 802.11ac throughput testing note

802.11ax: dense efficiency, not peak

Problem: 200 phones, one small frame at a time.



- **OFDMA:** frequency RUs
- **UL + DL MU-MIMO**
- **1024-QAM:** 10 bits/symbol
- **BSS coloring:** ignore faint cells
- **TWT:** the AP schedules naps

The design target moved: not a bigger single-link headline, but high efficiency when many stations share dense airspace.

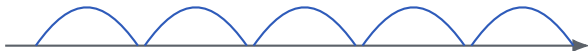
HE = High Efficiency Time UL = uplink foreign traffic RU = Resource Unit BSS = Basic Service Set TWT = Target Wake coloring = a per-cell tag on frames, so neighbors can talk over faint

Cisco 802.11ax white paper; IEEE Std 802.11

HE OFDM: four times narrower subcarriers

802.11a/g/n/ac

312.5 kHz spacing · 3.2 μ s useful symbol



802.11ax HE

78.125 kHz spacing · 12.8 μ s useful symbol



same 20 MHz channel, 4× more FFT points (conceptual, not a mask)

- Longer symbol → guard interval is relatively cheaper
- More subcarriers → finer-grained resource units
- Guard interval adapts: 0.8 / 1.6 / 3.2 μ s options

Narrower subcarriers are what make small OFDMA resource units — and outdoor-tolerant symbols — possible at all.

[Cisco 802.11ax white paper \(HE OFDM numerology\)](#)

Wi-Fi 6E adds spectrum, not a waveform

2.4 GHz

range + legacy

5 GHz

capacity, mature

6 GHz

clean wide channels

same HE PHY/MAC toolbox — new operating spectrum and channel plan

- 6E = 802.11ax certified into 6 GHz
- No legacy stations → no rent to the installed base
- Channel availability varies by country

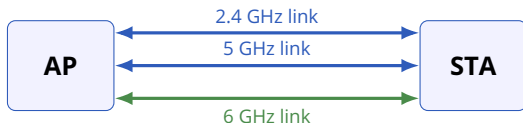
Q: Is Wi-Fi 6E a new generation?

A: No. A band extension is not a new IEEE amendment. The waveform is Wi-Fi 6's; the gift is clean, wide, uncontested spectrum.

[IEEE Std 802.11; Cisco Wi-Fi 6/6E materials](#)

802.11be: several links act as one

Problem: one association = one band; a busy band jails every retry.



multi-link operation: one association, several coordinated radios

- **320 MHz** (6 GHz only)
- **≥30 Gb/s** required mode
- **4096-QAM:** 12 bits, high SNR
- **Preamble puncturing:** skip busy MHz

MLO spends the last resource dimension: links themselves — throughput, balancing, redundancy, or latency, per traffic.

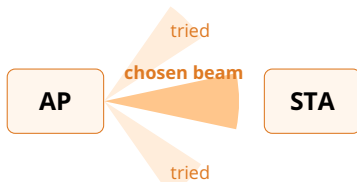
EHT = Extremely High Throughput
≥30 Gb/s is measured

MAC data SAP = MAC data service access point, where

IEEE Std 802.11be-2024

802.11ad: enormous bandwidth at 60 GHz

2.16 GHz channel **4.62 Gb/s** SC max rate **room-scale** range



severe path loss → the link closes only with aimed antenna gain
beam training: sweep sectors, keep the winner (conceptual)

- **Control PHY:** robust spread BPSK
- **Single-carrier PHY:** 4.62 Gb/s workhorse
- **OFDM PHY:** ≈ 6.8 Gb/s option, later obsoleted

Rate from raw bandwidth, not dense modulation. The price:
walls end the link, and every link must first be aimed.

Keysight 802.11ad concepts; CWNP 802.11 alternate PHYs white paper; IEEE Std 802.11

802.11ay: a wider, spatial 60 GHz system

802.11ad

one 2.16 GHz channel
directional single link



802.11ay

channel bonding + MIMO
 ≥ 20 Gb/s required mode

- Bond / aggregate 2.16 GHz channels
- Add MIMO and better beamforming
- Target longer-range outdoor links

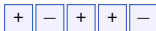
Q: Who buys a radio that cannot cross a wall?

A: Anyone replacing a cable: docking, VR tethers, and — at 11ay scale — fixed wireless access and street-level backhaul.

60 GHz evolved from cable replacement toward high-capacity access and backhaul.

Four waveform ideas explain the PHY story

DSSS / CCK



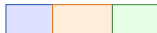
spread symbols across chips

OFDM



parallel orthogonal tones

OFDMA



assign tone groups to users

60 GHz SC



very wide directional carrier

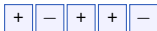
Sketches are conceptual, not calibrated time-domain or spectrum plots.

Waveform choice follows the channel: interference, delay spread, bandwidth, mobility, and hardware constraints.

IEEE Std 802.11; Cisco 802.11ax white paper; Keysight 802.11ad concepts

Four waveform ideas explain the PHY story

DSSS / CCK



spread symbols across chips

802.11b

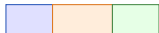
OFDM



parallel orthogonal tones

802.11a/g/n/ac

OFDMA



assign tone groups to users

802.11ax/be

60 GHz SC



very wide directional carrier

802.11ad/ay

Sketches are conceptual, not calibrated time-domain or spectrum plots.

Waveform choice follows the channel: interference, delay spread, bandwidth, mobility, and hardware constraints.

IEEE Std 802.11; Cisco 802.11ax white paper; Keysight 802.11ad concepts

The mainstream branch, b to be

802.11	year	GHz	MHz	key PHY idea	headline max
b	1999	2.4	≈ 22	CCK	11 Mb/s
a	1999	5	20	OFDM	54 Mb/s
g	2003	2.4	20	OFDM	54 Mb/s
n	2009	2.4/5	20/40	OFDM + MIMO	600 Mb/s
ac	2013	5	20–160	OFDM + MU-MIMO	6.9 Gb/s
ax	2021	2.4/5/6*	20–160	OFDMA + MU-MIMO	9.6 Gb/s
be	2024	2.4/5/6	20–320	EHT + MLO	≥ 30 Gb/s [†]

Years are amendment approval; 802.11be published July 2025. * 6 GHz operation is in 802.11ax-2021; availability depends on national regulation (Wi-Fi 6E). [†] Required mode at the MAC data SAP; peak PHY rate ≈ 46 Gb/s.

Maxima are comparison anchors, not promised throughput.
Standards allow more than typical devices ever implement.

[IEEE 802.11 timeline](#); [IEEE Std 802.11](#); [IEEE Std 802.11be-2024](#); [Cisco](#)

A PHY rate is a product of optimism

$$\text{Rate} \propto \text{bandwidth} \times \text{bits/symbol} \times \text{code rate} \times \text{streams}$$

then subtract, from radio up to application:

**preamble +
pilots**

airtime overhead

**contention +
ACK**

MAC overhead

retries
channel loss

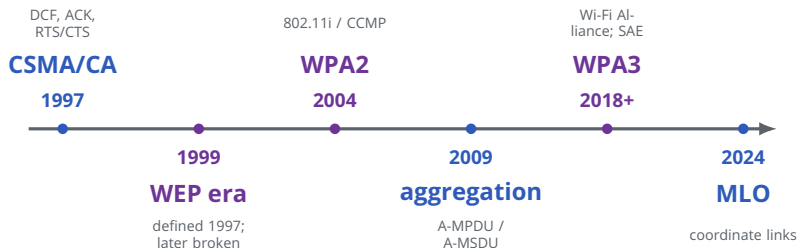
TCP / app
protocol overhead

Walking away from the AP, which factor on the top line fails first?

Application throughput is always lower — and depends on distance, interference, client count, and scheduler behavior.

Cisco, 802.11ac wireless throughput testing and validation

MAC and security evolved alongside



PHY innovation changes what is possible; MAC innovation decides who gets airtime — and how safely.

DCF = Distributed Coordination Function WEP/WPA = Wired Equivalent Privacy / Wi-Fi Protected Access CCMP = AES-based encryption of 802.11i SAE = Simultaneous Authentication of Equals

[IEEE Std 802.11; Wi-Fi Alliance security overview; IEEE Std 802.11be](#)

Wi-Fi and cellular optimize different edges

Wi-Fi

- Unlicensed spectrum
- You own the AP
- Contention-based access
- Very wide local channels
- Rapid device ecosystem

Cellular

- Licensed, managed spectrum
- Operator control
- Scheduled access
- Wide-area mobility
- Integrated service guarantees

Q: So which one wins?

A: Wrong question. Future edge systems choose, combine, and hand off across both — campus offload, AR/VR, industrial links, fixed wireless access, and edge computing already do.

IEEE GEPS, IoT and smart cities perspective

Read any generation in five questions

- 1 What spectrum does it use?
- 2 What does one PPDU look like?
- 3 How are users separated?
- 4 What channel conditions does it assume?
- 5 Which bottleneck moved next?

b → a/g → n → ac → ax → be (and ad → ay, off to the side)

Question 5's answers were the six rows of our opening table — each generation spent the next dimension.

Homework for your intuition: run the five questions on 802.11bn, the amendment being written right now.